

AI-assisted Meter Reading Validation RPA

Project summary

Repository root PDF

UiPath-visible automation with Python validation, SQLite history, controlled AI image extraction, and reviewed customer communication.

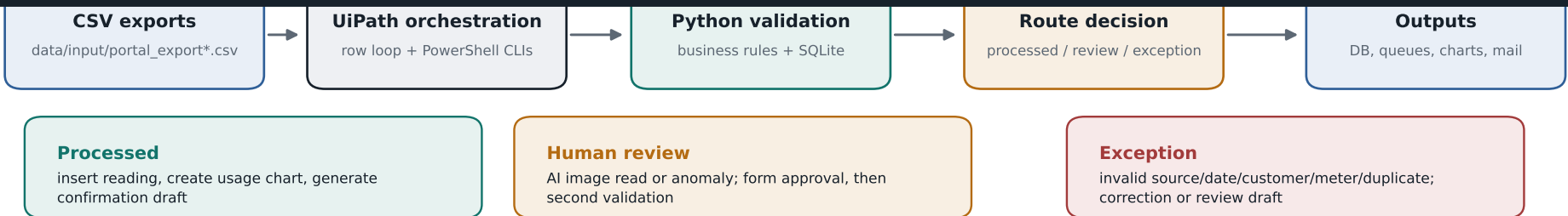
What It Does

- Automates a recurring energy-company meter reading validation process from portal/mobile CSV exports.
- Checks customer, meter, date, source, duplicate, reading, and consumption rules against local SQLite history.
- Inserts only clean, non-image readings automatically; uncertain records become exceptions or human review items.
- Uses GPT vision for photo-based readings only as a reviewer aid, then revalidates the human-approved value before insert.

Architecture And Key Components

- UiPath rpa/Main.xaml: scans data/input/*.csv, parses rows, calls Python CLIs through PowerShell, shows review forms, sends Gmail, deletes processed CSVs.
- Python CLI layer: validate_submission_cli.py, insert_from_validation_cli.py, extract_meter_image_cli.py, insert_reviewed_reading_cli.py, lookup_customer_email_cli.py, generate_ai_mail_draft_cli.py.
- meter_services package: contracts, validation, database inserts, chart generation, mail draft contracts, and run summary reporting.
- Data layer: data/history/master.db with customers, meters, meter_reading_history, and customer_contacts.

End-to-End Workflow



Validation And Safety Rules

- Allowed sources: portal, mobile_app, app_photo; reading_date must be valid YYYY-MM-DD and not future-dated.
- Customer and meter must exist, be active, and match ownership; one monthly reading per meter is allowed.
- Reading must be numeric, non-negative, and not lower than the latest previous reading.
- Consumption above 5000, image submissions, inactive records, and other uncertain cases go to human review.

Tech Stack

RPA	UiPath Studio 26; System, Form, Mail, Python, Excel activities
Python	pandas, matplotlib, OpenAI SDK, JSON CLI contracts
Storage	SQLite master.db plus CSV/TXT/PNG outputs
Secrets	.env / OPENAI_API_KEY for GPT calls

Repo State And Outputs

- Seed DB: 2 customers, 3 meters, 7 history rows, 2 contacts.
- Baseline inputs: 4 CSVs covering clean, image-review, and invalid-source cases.
- Outputs: processed_readings, exceptions, human_review_queue, mail_drafts, summary_report, charts.

Core principle: deterministic business validation controls database updates; AI assists extraction and drafting but does not bypass human review or revalidation.

Source: inspected repo docs, scripts, UiPath project, and SQLite schema